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Three-Dimensional Reconstruction of Dilute Bubbly Flow Field with Light-Field Images Based on Deep Learning Method

Hongyi Wang, Hongyu Wang, Xinjun Zhu, Limei Song, Qinghua Guo, Feng Dong

***Abstract*—The three-dimensional reconstruction of bubble flow field is of great significance to study the motion of gas-liquid two phase flow. Combined with the abundant information of bubbles in the light field images, a fast and accurate 3D reconstruction method for dilute bubbly flow based on the deep learning algorithm DIF-LeNet (Double Information Fusion with LeNet5 net) is proposed in this work. The calibration method and bubble segmentation algorithm of light field image for data processing is detailed. DIF-LeNet realizes the fusion of different dimension data and regression prediction. By DIF-LeNet, bubble depth could be predicted by fusing the refocused bubble image with the focal distance of the refocused image. Then, the three-dimensional bubble field could be reconstructed with the bubble depth, bubble center and bubble diameter. Comparing with the reconstruction method based on statistics and LeNet5, the reconstruction accuracy and speed of the proposed method based on DIF-LeNet are improved. Especially for the bubble without focused image in the refocused image sequence, the effect of 3D reconstruction based on DIF-LeNet is much prominent.**

***Index Terms*—Gas-liquid two phase flow, Bubble flow, Light field image, 3D reconstruction**

I. Introduction

BUBBLY flow is a common gas-liquid two phase flow pattern, which is widely exited in the areas of water conservancy, petroleum, chemical industry, nuclear industry, and so on [1-2]. Bubble characteristics measurement is important for improving production efficiency and optimizing product quality in these fields. Reconstruct the 3D field of bubble flow is helpful in better understanding bubble distribution and bubble motion.

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H. Wang, H. Wang, X. Zhu and L. Song are with the Key Laboratory of Advanced Electrical Engineering and Energy Technology, Tiangong University, TianJin 300387, China. (e-mail: wanghongyi@tjpu.edu.cn, 443780101@qq.com, xinjunzhu@tjpu.edu.cn, songlimei@tjpu.edu.cn). Q. Guo is with School of Electrical, Computer and Tele communications Engineering, University of Wollongong, Wollongong, NSW2500, Australia. (e-mail: qguo@uow.edu.au). D. Feng is with Tianjin Key Laboratory of Process Measurement and Control, School of Electrical and Information Engineering, Tianjin University, Tianjin, 300072, China. (e-mail: fdong@tju.edu.cn)

The 3D reconstruction method based on image is widely accepted because of its intuitive and high resolution. According to the needed images number, 3D reconstruction methods can be divided into two kinds: reconstruction based on single image and reconstruction based on multi-view images. Shape from shading (SFS) and shape from texture (SFT) are two typical single-image based methods for 3D reconstruction, which are mainly based on the image features of shadow, texture, focal distance and so on. The SFS method was proposed in 1970 by Horn [3] and a nonlinear partial differential equation (the shading equation) which shows the relationship between the brightness of each pixel in 2D image and the normal, the reflectivity and the illumination direction of the corresponding 3D point was given. SFS has already been used to reconstruct the surfaces of various types of objects quickly and accurately [4]. The SFT was proposed by Wiktin [5]. Sun [6] has used SFT and improved genetic algorithm to reconstruct the 3D texture image. Although it is simple and fast of 3D reconstruct method based on single image, the reconstruction accuracy and quality are low. Many methods of 3D reconstruction based on multi-view images were proposed, such as shape from focus (SFF) [7-8], shape from defocus (SFD) [9-10], photometric stereo (PS), structure from motion (SFM), and so on. These methods could greatly improve the accuracy and integrity of reconstruction. SFF and SFD are geometrical optics measurement methods, which restore the depth information of the object for 3D reconstruction based on the image sharpness, camera focal distance and aperture. PS method obtains object multi-images through a plurality of non-collinear light sources and gets the surface normal vector direction by establishing a set of brightness equations of different images to reconstruct the object shape and 3D model. Lin [11] described a novel closed generalized bas-relief (GBR) solution in the application of 3D surface detection and solved the reconstruction problems caused by shadow interference and other interferences by PS method. Xie [12] proposed a multi-view photometric stereo fusion algorithm for 3D movie special effects production, which achieves higher surface normal accuracy and better 3D reconstruction quality than traditional reconstruction methods. The SFM method detects the matched feature points in a set of un-calibrated images and restores the camera parameters and 3D information of the object through these feature points. Li [13] proposed the SFM method by combining surface points and dynamic structure to obtain complete 3D shape of object.

For gas-liquid two phase flow, several methods were studied to reconstruct the flow field. Tian [14] used ultrasonic tomography technology to analyze the gas-liquid two-phase flow, which enhanced the edge effect of the image and improved the accuracy of image reconstruction by setting dual thresholds. Tan [15] realized the measurement of bubble distribution by fusing ultrasonic attenuation and flight time. Jia [16] proposed the burst superimposed wave as a novel ultrasonic excitation, which achieved the measure of the particle size distribution in water. However, the ultrasonic measurement method is easily affected by the impurities in the water. Tian [17] used a DC motor to drive a small propeller in the water tank to generate bubbles, used the CCD sensor to obtain the image of bubbles, and finally used the holographic imaging measurement method to achieve the three-dimensional reconstruction of the bubbles. This method uses the laser as the light source, which leads to the complexity and high manufacturing cost of the equipment. Zhang [18] used one camera and four mirrors to take the images of the bubbles in the water which achieves the 3D reconstruction and tracking of single bubble. Although this method solves the problem that the single camera can only obtain the single angle information, but it also brings the problems of low matching accuracy and complex matching algorithms, which makes online measurement particularly difficult.

In recent years, the light field camera has been used to reconstruct the three-dimensional field, which benefits from the fact that the light field image can reflect the depth information of the target object. The light field was first proposed by Gershun [19] in 1963. This research regarded light as a geometric beam carrying energy, which represented the transmission of light radiation energy in 3D space. Lumsdaine [20] proposed the second-generation light field camera in 2009, which increased the imaging resolution from 625*434 to 3288*2192, and provided a hardware infrastructure for high-precision 3D measurement. The light field camera inserts a microlens array between the CCD sensor and the main lens, and light enters the microlens array through the main lens. Each microlens measures the total amount of light by each light incident at that position. Re-integrate the light in different positions to calculate the images which focused at different depths. Iwane [21-22] introduced how to use a light field camera to generate a volume 3D image. Zhao [23] used the light field camera to collect the light field information of 3D flame and reconstructed the temperature distribution of the 3D flame. Sun [24] observed the flame with the light field camera and realized the three-dimensional reconstruction and measurement of the flame temperature field by calculating the radiation intensity of the light. There are also some works related to 3D reconstruction of gas liquid flow filed. Cao [25-26] reconstructed the 3D particle field based on a light field camera by multiplicative algebraic reconstruction technique. Although researchers have used light field imaging technology to measure the corresponding research objects in many fields such as solid phase, liquid-solid two-phase flow and flame, only Li [27] studied the 3D measurement method of bubble flow in gas-liquid two-phase flow by using a light field camera with its total focused image and refocused image. In Li's work, a statistical method by calculating and comparing the sharpness of pixels of all the refocused images is employed to acquire the bubble depth. It is of high 3D reconstruction accuracy for the bubble with focused image in the refocused image sequence, but time-consuming. Another important issue is that the view field of the light field image is slightly changed with the focal distance of each microlens. There might be some edge bubbles move into or move out of the refocused image and have no focused image. And the bubble without focused image is hard to acquire the accurate depth by the statistical method.

With the development of deep learning technology, CNN has been used in the reconstruction of multi-phase flow. Tan [28] used CNN to reconstruct the gas-liquid two-phase flow field based on electrical resistance tomography (ERT), and the reconstruction accuracy and speed are better than traditional methods. Lei [29] used deep learning-based inversion method to reconstruct gas-solid two-phase images of electrical capacitance tomography (ECT), which reduces the reconstruction artifacts and deformations and improved the imaging quality. Haas [30] proposed the BubCNN, which employed a faster region-based CNN (RCNN) detector to locate bubbles and a shape regression CNN to predict bubble shape parameters. To predict the target parameter, CNN can automatically extract a variety of effective features from the training data, which could greatly improve the ability of target recognition. Thus, the CNN presents an outstanding performance in the field of digital image reconstruction. The key point of the CNN application is how to build a suitable network structure based on the characteristics of the sensor data to get the target.

Based on the problems above, this paper studies the method of 3D reconstruction with light field image. Considering that the depth of bubbles is related to the sharpness in the refocused image taken by the light field camera, an improved deep learning network called Double Information Fusion with LeNet5 (DIF-LeNet) is proposed in this paper. The DIF-LeNet could predict the depth of bubbles during the reconstruction of the three-dimensional bubble field. The acquisition and calibration of light field image, the preparation method of training data, the network structure of DIF-LeNet, the results of 3D reconstruction and error analysis are detailed in this paper. The deep learning network of DIF-LeNet is an improvement of LeNet5 [31]. It realizes the fusion of input data with different dimension and changes the original model into regression model. The DIF-LeNet predicts the bubble depth only by inputting the bubble image segmented from any refocused image and the focal distance of the refocused image. Furthermore, combining with the bubble center and radius computed by digital image processing method, the three-dimensional bubble flow field could be reconstructed. Compared with other 3D reconstruction algorithms based on light field camera, the proposed 3D reconstruction method has higher reconstruction accuracy and faster reconstruction speed.

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II. IMAGE DATA ACQUISITION

A. Experimental equipment and calibration

1) Experimental equipment: The schematic view of experimental setup is shown in Fig. 1. The bubbles are generated from nozzles at the bottom of a water tank. The diameter of the nozzles is between 0.3 to 1.8 mm. The size of the water tank is 17cm*17cm*100cm, which is made of acrylic material for the bubble visualization purpose. A LED backlight is used for illumination.

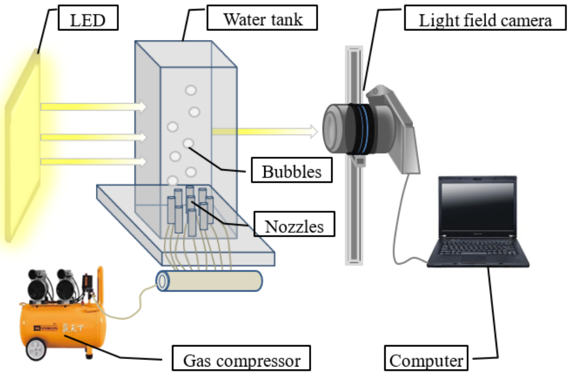


Fig. 1. Schematic view of the experimental setup

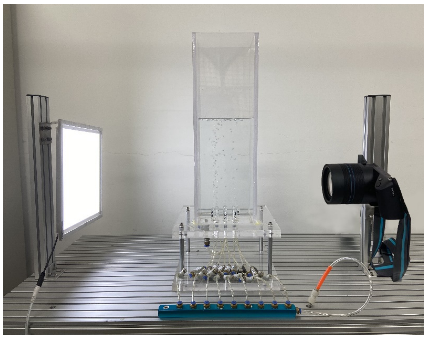
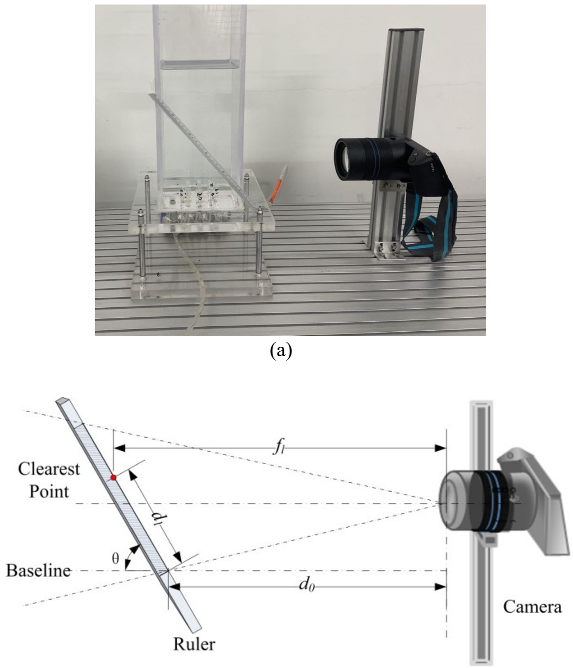


Fig. 2. Experimental system of bubble image acquisition

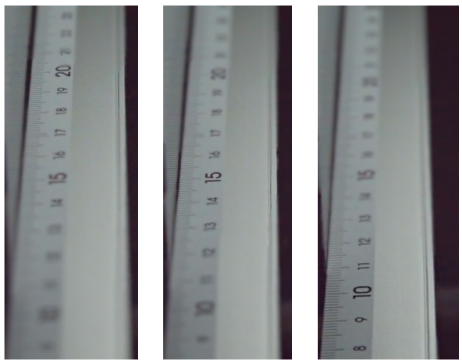
A LYTRO light field camera with the shutter speed of 1/4000s is employed to record bubble images. There is a microlens array between the main lens and the image sensor in the light field camera. After passing through the main lens of the light field camera, light is projected to the microlens array and imaged again. The pixel value after the microlens array not only records the light intensity information, but also records the direction information of each light beam relative to each microlens position. Therefore, all light passing through the main lens can be recorded by the microlens array and the photoelectric sensor, and then determined by the Fourier slice in the later stage. The theory and the light field imaging algorithm can refocus the light by retracing it [32]. When the image was taken by a light field camera, a total focused image and a refocused image sequence will be obtained. In this experiment, 390 refocused images and 1 total focused image would be acquired by the light field camera every time.

2) Calibration: In order to calibrate the focal distance of each focused image, a calibration experiment is performed before collecting bubble images. Set the calibration ruler in the measured space to an angle θ with the horizontal plane, as shown in Fig. 3(a). The focused images of the experimental calibration acquired by light field camera are shown in Fig. 4. Fig. 4(a), (b), and (c) are the no.102, no.159 and no.337 images in the refocused image sequence respectively.



(b)

Fig. 3. Calibration experiment. (a) Experiment site of calibration; (b) Geometric relationship between focal distance and ruler position.



(a) (b) (c)

Fig. 4. Calibration images of the ruler. (a) refocused image 1; (b) refocused image 2; (c) refocused image 3.

It is obviously that the clear position in the ruler is related to the focal distance of the refocused image. In order to improve the accuracy of clear point positioning, the sharpness of each graduated line on the calibration ruler in every refocused image could be evaluated by the sharpness evaluation function:

$$S = \frac{1}{N}\sum_{j=1}^{N}\sum_{k=1}^{8}\left|\frac{\Delta F_{jk}}{\Delta D_{jk}}\right| \tag{1}$$

where, S is sharpness of image pixel, N is the pixel number of the graduated line, j is the j-th pixel of graduated line, k is the k-th pixel of the 8-neighourhood for pixel j, ΔF_{jk} is the changed gray value between pixel j and its k-th pixel of the 8-neighourhood, ΔD_{jk} is the distance between pixel j and its k-th pixel of the 8-neighourhood.

The lager the S is, the clearer the graduated line is. The clearest position of the ruler in every refocused image could be acquired by finding the clearest pixel. The Geometric relationship between the focal distance and the ruler position is shown in Fig. 3(b). Then, the focal distance f_l of the refocused image could be calculated by formula (2).

$$f_l = d_0 + d_l \cdot cos\theta \qquad (2)$$

where, d_0 is the horizontal distance between the camera and the baseline point, d_l is the distance between the clearest point and baseline point, θ is the angle between the ruler and the baseline.

Since the minimum scale of the calibration ruler is 1mm, not the focal distance of every refocused image is at the graduated line. It is necessary to calibrate the refocused images which are not focused at the graduated line by linear interpolation. If the clearest graduated line appears in the refocused image, then marks focal distance of the refocused image value as f_l by formula (2). Otherwise, select the two refocused images p and q which the adjacent graduated lines are focused, calculate the distance between the adjacent graduated lines and the number of middle refocused images between p and q, evenly distribute the distance to the middle refocused images. The formula of the linear interpolation method is

$$f_l^{p+i+1} = f_l^{p+i} + \frac{f_l{}^q - f_l{}^p}{|q-p|+1} \qquad (3)$$

where, $f_l{}^q$ and $f_l{}^p$ are the focal distance of the two refocused images, i is the i-th image of the middle refocused images.

Thus, the focal distance of every refocused image could be calibrated by the above method. A querying dictionary which is about the serial number of the refocused image and its focal distance could be established. An example of the querying dictionary for this work is shown in TABLE I.

TABLE I
QUERYING DICTIONARY

Focal distance (cm)	···	13.000	12.980	12.960	…	9.524	9.516	9.508	9.500	…
Serial number	···	180	181	182	…	300	301	302	303	…

In this work, the angle θ between the calibration ruler and the horizontal plane in Fig. 3 is 30°. The minimum scale of the calibration ruler is 0.1cm as shown in Fig. 4. After the linear interpolation processing, the measurement error range of the celebrate focal distance is (-0.016cm~0.016cm).

It is worth noting that in the case of fixed experimental device, the bubble light field images could be obtained repeatedly after a calibration experiment. When the position of the experimental device changes, it is necessary to conduct the calibration experiment again and make the data set according to the new calibration results. No matter how many experiments are carried out, as long as the correct calibration is carried out, the experimental data could be integrated and applied to establish the model.

B. Data acquisition

1) Single bubble image extraction: Bubble images are recorded with the experimental setup of Fig. 2. An example of the light field refocused bubble image is shown in Fig. 5. As can be seen, some bubbles are clear and some bubbles are blurred because of the different depth of bubble in the tank.

For further training in the depth computation network, image data of single bubbles should be prepared. Image processing method of image cutting and matching is necessary. But it is impossible to find the same bubble in different light field images through the absolute position. Because both the bubble position and the image field in the refocused image sequence are slightly different. This is caused by the different viewing angles of the sub-aperture of the main lens in the light field camera. And then another problem follows, some bubbles may not appear in every refocused image. As shown in Fig. 5, bubbles drown out by circles are gradually appeared in the image sequence. Thus, the typical labeling method could not get the matching bubble series accurately from the refocused image sequence.

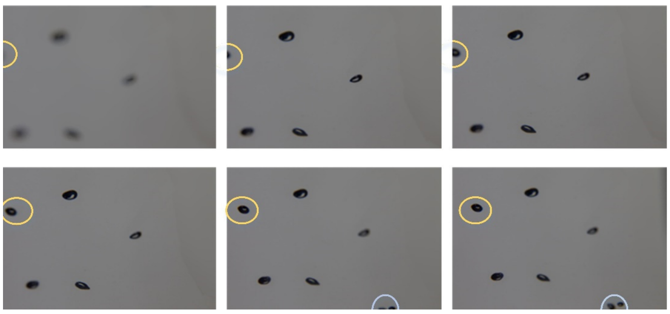


Fig. 5. Refocused images of bubble.

In order to solve this problem, this paper improves the labeling matching method, which could realize the extraction and matching of the target bubbles in the refocused image sequence. The improved labeling matching method is shown in Fig. 6.

The key point of this improved method is to determine whether there are bubbles that do not appear in all the refocused images. It can be observed that the bubble might enter or run out of the field of view if the image edge contacts with the bubble. For the camera in this work, the image sequence follows the rule that the field of view gradually moves to the right and up and the changes of the field of view between two adjacent images is slightly. If the bottom or left edge is in contact with the bubble, the bubble will enter into the field of view. If the top or right edge is in contact with the bubble, the bubble will run out of the field of view. When a bubble enters into the image, a new label will be added to the new bubble. On the other hand, when a bubble moves out of the image, the label of the out bubble will be kept and not allowed to use for the other bubbles.

An example of the extraction bubbles from the refocused image sequence is shown in Fig. 7. The size of the single bubble image is 96*96 pixels in this work.

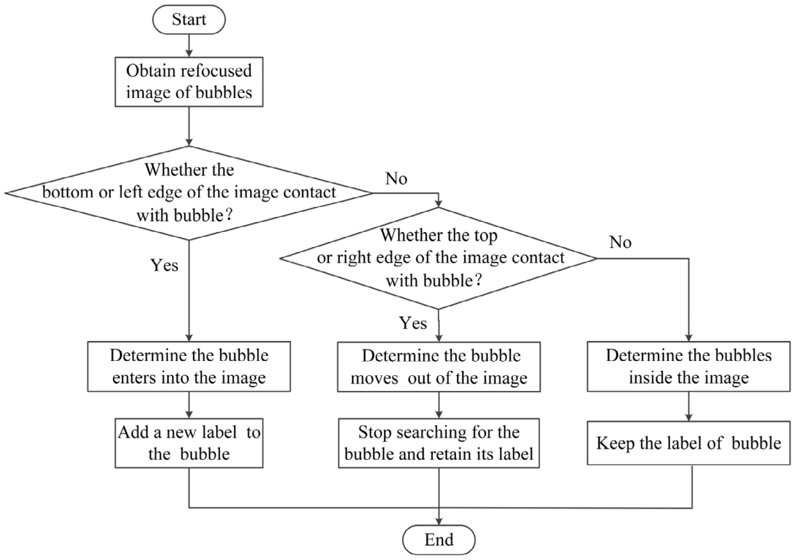


Fig. 6. Improved labeling matching method for single bubble extraction

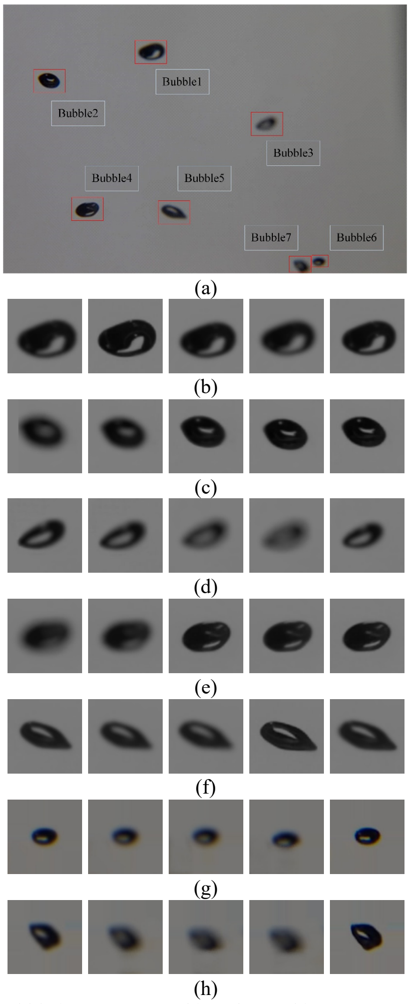


Fig. 7. Single bubble image extracted in refocused image sequence. (a) labeled bubble in refocused image; (b)bubble 1, depth=11.3cm; (c) bubble 2, depth=12.3cm; (d) bubble 3, depth=11.9cm; (e) bubble 4, depth=12.0cm; (f) bubble 5, depth=12.2cm; (g) bubble 6, depth=12.5cm; (h) bubble 7, depth=11.7cm.

2) Bubble depth acquisition: According to the principle of light field imaging, it can be known that the object only has one focus position in all refocused images. In other words, each bubble corresponds to a certain depth even though it appears in so many refocused images. Based on this theory, the ground truth depth of the bubble and the focus sign of every refocused image can be obtained by the following four steps:

(i) Establish the querying dictionary Z which could reflect the relation between the serial number of the refocused image and its focal distance by calibration, as described in section II-A.

(ii) Segment the single bubble from the refocused image sequence by using the labeling matching method to rebuild a new image sequence, and record the refocused image number R_n to every segmented image.

(iii) Calculate the sharpness of bubble g in every segmented bubble image by formula (1). And then find the focused bubble image with the maximum sharpness. Record the focus sign of the segmented bubble image as 1 if the bubble is focused in this image; otherwise, recorded as 0. Find and record the refocused image number R_{ng} of bubble g.

(iv) Query the focused distance of the refocused image R_{ng} from the querying dictionary Z. The ground truth depth of bubble g is the focused distance of the refocused image R_{ng}, which is marked as h_g. As shown in Fig. 7, the bubble ground truth depths of these bubbles are shown in the title.

III. Deep Learning Model for Bubble Depth

In the traditional 3D reconstruction method based on light field image, the sharpness of all refocused images (390 in this work) of bubbles should be calculated by formula (1). The actual depth of bubble is related to the focal distance of the refocused image. By fusing with parameters of bubble size and position, the three-dimensional reconstruction of bubble field could be obtained.

The classical convolution neural network (CNN) could obtain the bubble depth information by classifying the single bubble image to find the focused image. The LeNet5 is a classical convolution neural network for classification with single input and single output [21]. But this method could not compute the depth of bubble without bubble focused image. Meanwhile, the computational complexity of this method is very large.

Considering that the sharpness of bubble in the image is closely related to the focal distance of the light field camera microlens and the depth of the bubble. It is possible to set a model to predict the bubble depth information by the refocused bubble image and the focal distance of the light field camera microlens. In order to promoting the computational efficiency and reducing the computational complexity, an improved LeNet5 network with double information fusion (DIF-LeNet) is proposed in this work. By the deep learning algorithm of DIF-LeNet, the depth of all bubbles could be predicted by one piece of bubble refocused image and the focal distance of this image. The whole structure of the DIF-LeNet is shown in Fig. 8. The input, network structure and output of DIF-LeNet are all innovated and improved.

1) Two input layers are sent to DIF-LeNet. One is the single bubble image (2D data) segmented from the refocused image, and the other is the focal distance of the refocused image (1D data). The fusion of input data with different dimensions is realized by DIF-LeNet.

2) There are three convolutional neural networks in DIF-LeNet. CNN1 transform one-dimensional data to 2D data by convolution operation. CNN2 extracts the features of 2D image by multiple convolutions, pooling and batch normalization operations. CNN3 merges the operation results of CNN1 and CNN2 and extracts the secondary features from the two results.

3) Four full connection layers are designed at the end of CNN3 to transform the feature space, integrate information and calculate the output. The neuron cell in the last fully connected layer is adjusted to realize regression output instead of classification output. Through the full connection layers, all features will be fused again and the depth of bubble could be put out.

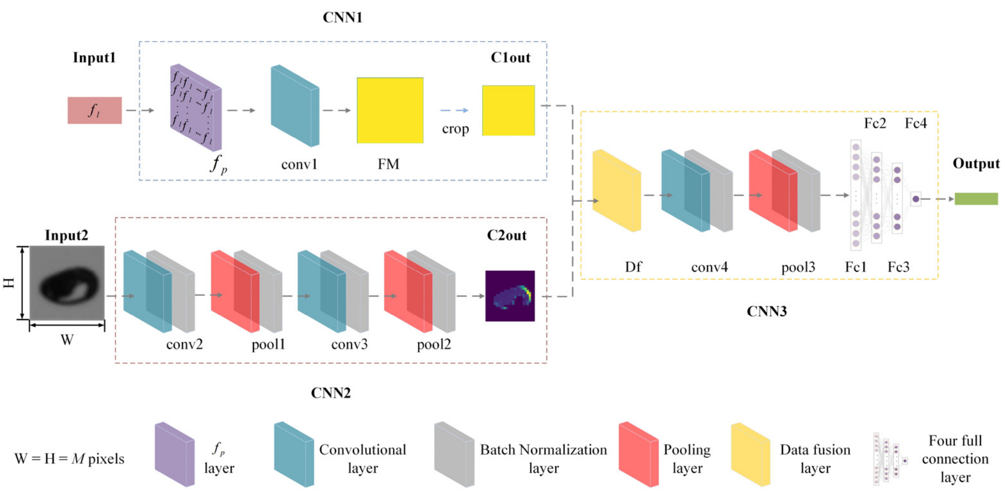


Fig. 8. Sketch of DIF-LeNet

A. CNN1 layer

The CNN1 is constructed by two convolution layers and one crop layer, as shown in Fig. 8.

Firstly, transform 1D data into 2D. The input of CNN1 is the focal distance of the refocused bubble image which is the input of CNN2. To fuse the focal distance data of CNN1 with the image data of CNN2, it is necessary to unify the data size of them. Considering that the image input into CNN2 is a matrix with the size of $M*M*3$, the focal distance input into CNN1 is transformed into a matrix with the size of $M*M*1$. The formula of the first convolution layer of CNN1 is:

$$f_P = \mathrm{E} \cdot f_l \tag{4}$$

where, f_l is the focal distance of the refocused bubble image, E is the convolution matrix with all elements of one and the size of E is $M*M$.

Secondly, a discrete convolution on f_P would be carried out. Considering that the elements of f_P are all the same, only one convolution operation is performed on f_P to extract the focal distance feature matrix FM.

Thirdly, crop the matrix FM to get the output $\mathrm{C1}_{\mathrm{out}}$ of CNN1. In order to fuse the results of CNN1 and CNN2, the matrix FM is cropped into the same size with the output of CNN2.

B. CNN2 layer

The network structure of CNN2 is a combination of convolution layer and pooling layer in a typical convolution neural network, as shown in Fig. 8. The obtained convolution images enhance the different characteristics of bubbles by CNN2 network, such as the bubble edge sharpness, bubble size, the image resolution, and so on. Part of the changing process of bubble image features in CNN2 is shown in Fig. 9. For special, the output of CNN2 is particularly sufficient in the expression of bubble edge information since the gradient value of bubble edge is a main feature of bubble sharpness. Although the features of CNN2 could express the bubble sharpness information, it is impossible to get the bubble depth only by image sharpness.

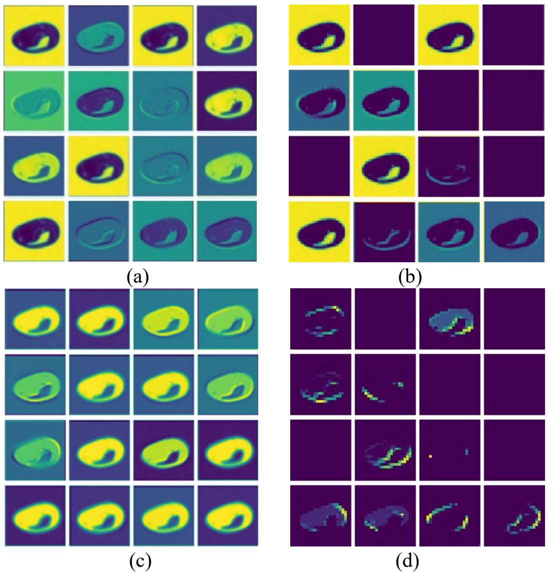


Fig. 9. Examples of feature map visualization in CNN2. (a) Feature map visualization of the first convolution result, (b) Feature map visualization of the first pooling result, (c) Feature map visualization of the second convolution result, (d)Feature map visualization of the second pooling result.

C. CNN3 layer

CNN3 is mainly composed of the data fusion layer, convolution layers and full connection layers, as shown in Fig. 8. It is noteworthy that the structure of data fusion layer and full connection layer is improved.

1) Data fusion layer

The main task of CNN3 is to fuse the output feature matrixes of CNN1 and CNN2. Considering that the number of channels of CNN2 is twice of CNN1 (as shown in TABLE II), a fusion regulation of formula (5) is used in the data fusion layer.

$$F(u,v,1:3) = [C1_{out}(u,v,c), C2_{out}(u,v,2c-1), C2_{out}(u,v,2c)] \quad (5)$$

where, $C1_{out}$ is the output feature matrix of CNN1 layer, $C2_{out}$ is the output feature matrix of CNN2 layer, (u,v,c) is the position in the feature matrix, c is the channel number, $c=\{1,2,\ldots,16\}$. Obviously, F is a three-dimensional matrix with overlaid by three 2D matrixes. The fusion regulation is shown in Fig. 10.

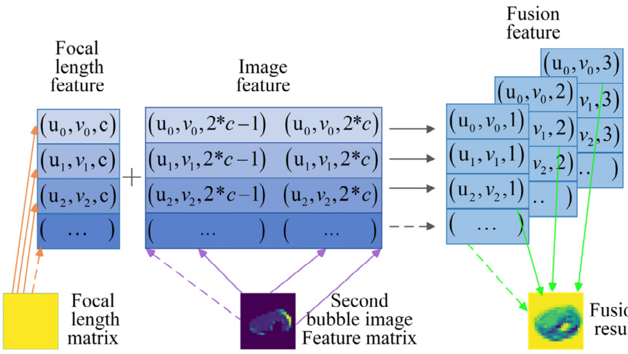


Fig. 10. Data fusion regulation

Fig. 11 shows the feature fusion results of $C1_{out}$ and $C2_{out}$ by CNN3. For the feature fusion results, more details are enhanced, the contrast and the pixel level of the bubble become clearer. Both the bubble edge and the textual of the bubble are enriched, which is helpful to predict bubble depth. This is because the output feature of CNN1 is added to the output image features of CNN2 with an appropriate weight ratio, which jointly determine the values of each pixel in the fused image. This principle is similar to the constitution of the image display color by the RGB color component and Alpha value.

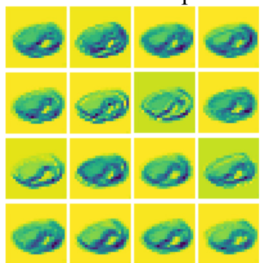


Fig. 11. Feature fusion results by CNN3

2) Full connection layer

The task of full connection layer of CNN3 is transforming the feature matrix to an ordered 1D eigenvector and realizing the regression prediction of bubble depth. Since the typical LeNet5 is a classification network, the structure of the full connection layer needed to be improved to the regression prediction in this work. Thus, the number of full connection layer is set to four.

The output of the first full connection layer is calculated by:

$$Fc1 = W1 * P3_{out} + B1 \quad (6)$$

which could be expanded to formula (7):

$$Fc1 = \begin{bmatrix} w1_{11} & w1_{12} & \cdots w1_{1m} \\ w1_{21} & w1_{22} & \cdots w1_{2m} \\ \vdots & \vdots & \vdots \\ w1_{n1} & w1_{n2} & \cdots w1_{nm} \end{bmatrix} * \begin{bmatrix} p3_{out1} \\ p3_{out2} \\ \vdots \\ p3_{outm} \end{bmatrix} + \begin{bmatrix} b1_1 \\ b1_2 \\ \vdots \\ b1_n \end{bmatrix} \quad (7)$$

where, $W1$ is the weight matrix of first full connection layer FC1 and $w1_{nm} \sim N(0, 0.1^2)$, $P3_{\mathrm{out}}$ is the output matrix of pool3 layer in CNN3, B is the bias matrix and $b1_n \sim U(0,1)$, n is the neuron cell number of the connection layer, m is the hidden layer number of the connection layer.

In the same way, the second and third full connection layers are computed by formula (8) and (9).

$$Fc2 = W2 * Fc1 + B2 \quad (8)$$

$$Fc3 = W3 * Fc2 + B3 \quad (9)$$

In this work, the neural cell number n of first three connection layers are 2048, 1000 and 500 respectively, the hidden layers m of first three connection layers are 128, 2048 and 100 respectively.

Then, the fourth full connection layer is used for regression output by formula (10).

$$Fc4 = W4 * Fc3 + B4 \quad (10)$$

Formula (10) could be expanded to formula (11).

$$Fc4 = [w4_1 \quad w4_2 \quad \cdots \quad w4_m\ ] * \begin{bmatrix} Fc3_1 \\ Fc3_2 \\ \vdots \\ Fc3_m \end{bmatrix} + b4 \quad (11)$$

where, $W4$ is the weight matrix of size 1*m of the fourth connection layer. There is only one hidden layer and the neuron number m=500 in the fourth connection layer.

The four connection layers realize the combination of all the information, and realize the transformation from classification network to regression network.

D. Parameter sets of the DIF-LeNet

The parameters of the DIF-LeNet proposed in this work are set as TABLE II.

TABLE II
DIF-LENET PARAMETER SETS

	Layer	Number of cores	Kernel size	Stride size	Padding (Y or N)	Output size
CNN1	Input1			1*1*1		
	f_P layer	/	/	/	/	96*96*1
	conv1	16	3*3*16	1	Y	96*96*16
CNN2	Input2			96*96*3		
	conv2	16	3*3*16	1	Y	96*96*16
	pool1	/	2*2*1	2	/	48*48*16
	conv3	32	3*3*32	1	Y	48*48*32
	pool2	/	2*2*1	2	/	24*24*32
CNN3	Df layer	/	/	/	/	24*24*48
	conv4	128	3*3*128	1	Y	24*24*128
	pool3	/	2*2*1	2	/	12*12*128
	Fc1	/	/	/	/	2048*1
	Fc2	/	/	/	/	1000*1
	Fc3	/	/	/	/	500*1
	Fc4	/	/	/	/	1*1
	Output layer		Output regression depth information			

Note: conv* stands for the *th convolutional layer; pool* stands for the *th pooling layer; Fc* stands for the *th fully connection layer layer.

IV. THREE-DIMENSION RECONSTRUCTION OF BUBBLE FIELD

A. Data sets for experiments

The experiment of 3D reconstruction is implemented with Tensorflow framework and calculated on a Geforce RTX 2070 with max-q design graphics card (NVIDIA). Three kinds of 3D reconstruction method (DIF-LeNet model, LeNet5 model and statistical method) are compared in this work.

For the network model, there are 2040 refocused light field images of single bubble acquired by the image processing method as introduced in section II-B. The depth of these bubbles, the focal distance and the focus sign of the refocused single bubble image are acquired by the method detail in section II. As the bubble generated in the middle area of the water tank, the range of bubble depth is from 10cm to 17cm in this experiment. According to the ratio of 9:1, all the collected bubble images were randomly divided into the training image set and testing image set. For DIF-LeNet model and LeNet5 model, the training image set and testing image set are the same. The percentage partition of training, testing and validation image sets is shown in Fig. 12. However, according to the input and output characteristics of the network structure, there are some differences of the data sets for different models. For the proposed method of DIF-LeNet, the data sets are combined of the segmented single bubble image, the focal distance of the corresponding refocused image and the bubble depth. The first two data are used as the input data and the last one is used as the target output. For LeNet5, the data sets are combined of the segmented single bubble image and the focus sign of each bubble image. The segmented single bubble image is used as the input data and the focus sign is used as the target output. That means bubble refocused images can be divided into the class of focused and unfocused by LeNet5 classification.

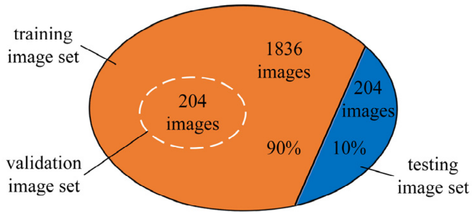


Fig. 12. The composition of data set for model establishment

In addition, to verify the 3D reconstruction result of bubble flow field, a bubble flow field image set composed of several refocused light field image sequences with multi-bubble are reserved, which are independent of the training image set and testing image set. To comply with algorithm requirements, all the images in the bubble flow field image set should also be calibrated and processed by the method detailed in section II. For better verification, the image sequence without focused bubble image will be removed from this set. Then, 50 refocused light field image sequences of single bubble collected from the reserved multi-bubble light field image sequences are prepared as 3D reconstruction verification image set.

B. Experiments of 3D bubble field reconstruction based on DIF-LeNet

The 3D bubble field reconstruction process is shown in Fig. 13.

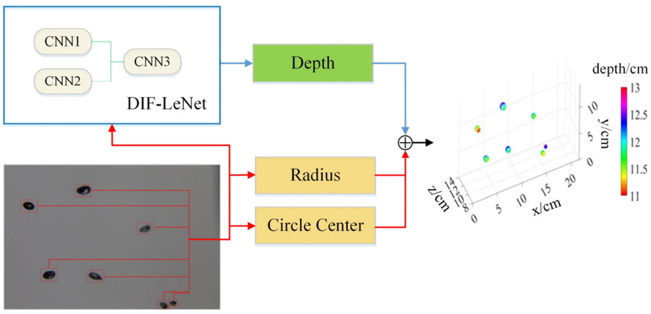


Fig. 13. The 3D reconstruction process of bubble field

On the premise of calibrating the world coordinates of bubble image, the 3D bubble field could be reconstructed based on a refocused image and its focal distance according to the following steps:

(a) Acquiring the focal distance of the refocused image according the calibration result.

(b) Segmenting single bubble from the refocused image.

(c) Predicting the depth of every bubble by DIF-LeNet model.

(d) Computing the equivalent radius and center of every bubble by digital image processing method [33].

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(e) Reconstructing the 3D bubble field with the 3D coordinates of bubbles. The bubbles are represented by spheres of bubble equivalent radius.

The depth of bubble is calculated by DIF-LeNet model detailed in section III. In this work, the root mean squared error (RMSE) is used as the loss function to evaluate the error of the model. The RMSE function is shown in formula (12).

$$\mathrm{RMSE} = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(y_{predict}^{i} - y_{true}^{i})^2} \tag{12}$$

where, m is the number of training image, $y_{predict}^{i}$ is the predicted bubble depth of the *i-th* image, y_{true}^{i} is the true depth of the *i-th* image.

The accuracy (ACC) used to evaluate the deviation of the model's predicted depth from the ground truth depth is defined by formula (13).

$$\mathrm{ACC} = \left(1 - \left|\frac{D_{predict} - D_{true}}{D_{true}}\right|\right) * 100\% \tag{13}$$

where, $D_{predict}$ is the predicted depth of the bubble, D_{true} is the ground truth depth of the bubble.

The loss of the DIF-LeNet model with batchsize 16 is shown in Fig. 14. As can be seen, the loss value tends to be stable after 50 iterations, which proves the robustness and astringency of the model. To ensure the better performance of the model, the number of iterations is increased to 1300.

The testing results of DIF-LeNet are shown in Fig. 15. As can be seen, the proposed method shows a good prediction effect on the depth of bubbles in the entire depth range. The relative error of the predicted depth is lower than 4.2% and the mean relative error is 1.5%. According to formula (12), the highest ACC, the lowest ACC and the mean ACC for validation and test are shown in TABLE III. All of the experimental results have proved the effectiveness of the proposed DIF-LeNet model.

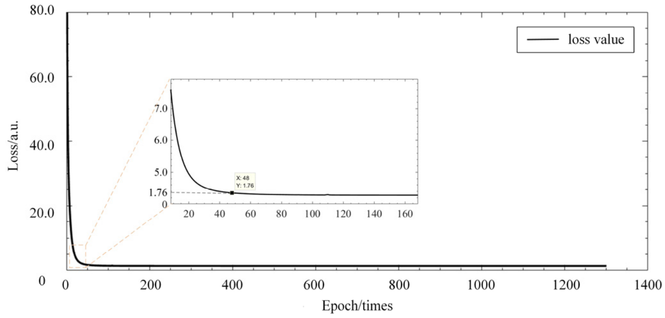

Fig. 14. Loss of the DIF-LeNet

TABLE III
ACCURACY OF THE PREDICTED BUBBLE DEPTH

	Sample number	The highest ACC	The lowest ACC	Mean ACC
validation	204	0.9927	0.9662	0.9886
Test	204	0.9912	0.9588	0.9855

Once the bubble depth model of DIF-LeNet was established, the 3D bubble flow field could be reconstructed based on any refocused light field image of multiple bubbles according to the process of Fig. 13. An example of 3D reconstruction of bubble flow field based on DIF-LeNet method is shown in Fig. 16. Fig. 16(a) is the original refocused light field image. Fig. 16(b), (c) and (d) are the 3D reconstruction results of bubble field viewed from different angle. The correlation between bubbles in the original image in Fig. 16(a) and the reconstructed field in Fig. 16(c) is shown by dotted lines. As can be seen, the distribution and size of the bubbles in the reconstructed field are well consistent with those in the refocused image.

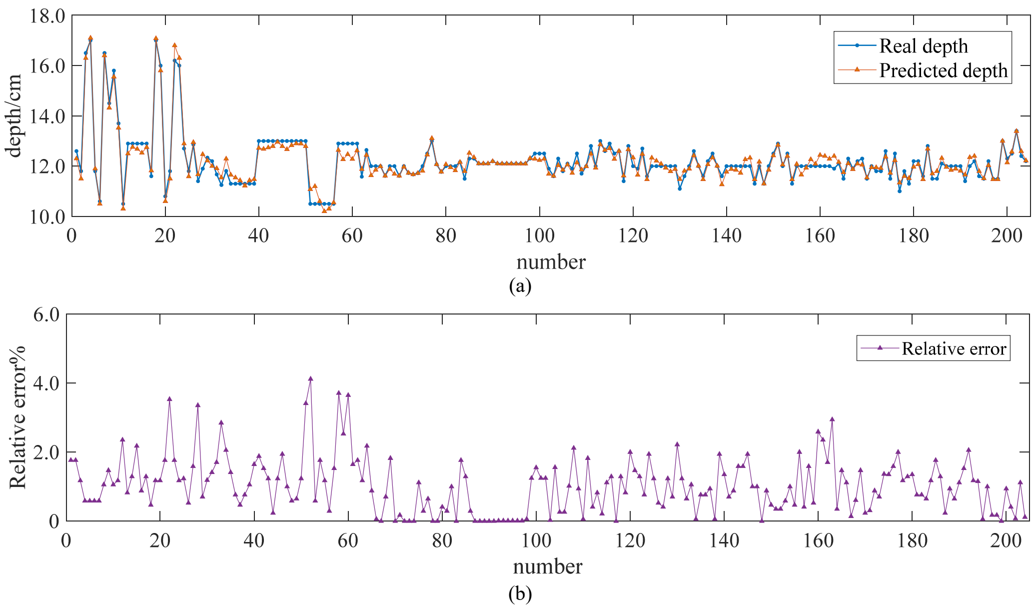

Fig. 14. The testing results of DIF-LeNet model. (a) Predicted bubble depth of the testing set; (b) Relative error distribution.

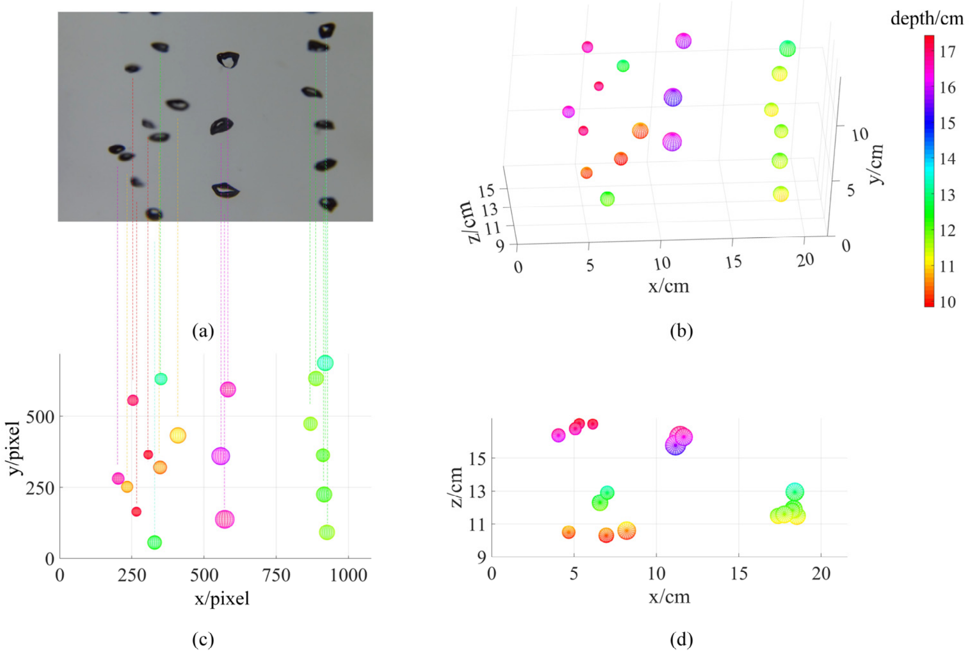

Fig. 16. Three-dimensional reconstruction result of bubble flow field.

C. Comparison of different reconstruction methods

The outstanding advantage of the reconstruction method proposed in this work is not only fast computing, but also could accurately locate the unfocused bubble in the image sequence. In order to further verify the effectiveness of the proposed method, the 3D reconstruction result based on the statistics method and LeNet5 classification method are employed for comparison. For statistics method, the bubble depth is found by computing the sharpness of the bubble in all refocused images and searching the focal distance of the bubble image with the largest sharpness. For LeNet5 classification method, the single bubble images are classified into focused image and un-focused images, and the bubble depth is corresponding to the focused image. During the 3D reconstruction process based on these two methods, the bubble size and center are calculated by the same digital image processing method as for DIF-LeNet.

To compare the 3D reconstruction results of the three methods mentioned above, it is necessary to choose and design a suitable data set that meets all the requirements of the three reconstruction methods. Considering that both the LeNet5-based 3D reconstruction method and the statistical-based 3D reconstruction method determine the bubble depth by finding the focused image in the refocused bubble image sequence, the image data set used for the reconstruction comparison should contain the complete refocused image sequence. Therefore, the 3D reconstruction verification image set introduced in section IV-A is employed here. Furthermore, in viewing of the possibility that the edge bubbles may have no focused image in the entire group of refocused images, we simulate this situation with the experimental data by removing some refocused images which contains focused bubbles from 3 groups of the refocused image sequences. As shown in Fig. 17(a), the focused image of the framed bubbles is removed from the refocused image sequence, and 380 images were remained in its refocused image sequence. Thus, this second-processed image set of bubble refocused light field image sequences is employed for 3D reconstruction to the three reconstruction methods.

The contrast of 3D reconstruction result of bubble field based on different methods is shown in Fig. 17. Fig. 17(a) is an image sample of multi-bubble refocused light field image selected randomly from the refocused light field image sequence. In Fig. 17(b), the reconstruction results of DIF-LeNet method are based on the refocused image of Fig. 17(a), the reconstruction results of LeNet5 method and statistical method are based on the refocused light field image sequence. As can be seen, the 3D position calculated by DIF-LeNet is closer to the real position than the other two methods. The standard deviation (StdDev), the root means square error (RMSE) and the mean absolute error (MAE) are used to measure the reconstruction error of the three methods about the bubbles in Fig. 17(a), as shown in Fig. 17(c). For DIF-LeNet method, the error in Fig. 17(c) is the mean reconstruction error with every refocused light field image in the refocused field image sequences. Compared with the other two methods, the proposed method has the lowest errors of bubble position. In particular, for the bubble without focused image in the whole refocused image sequence, the DIF-LeNet shows good prediction performance, but both statistics method and LeNet5 classification method are invalid.

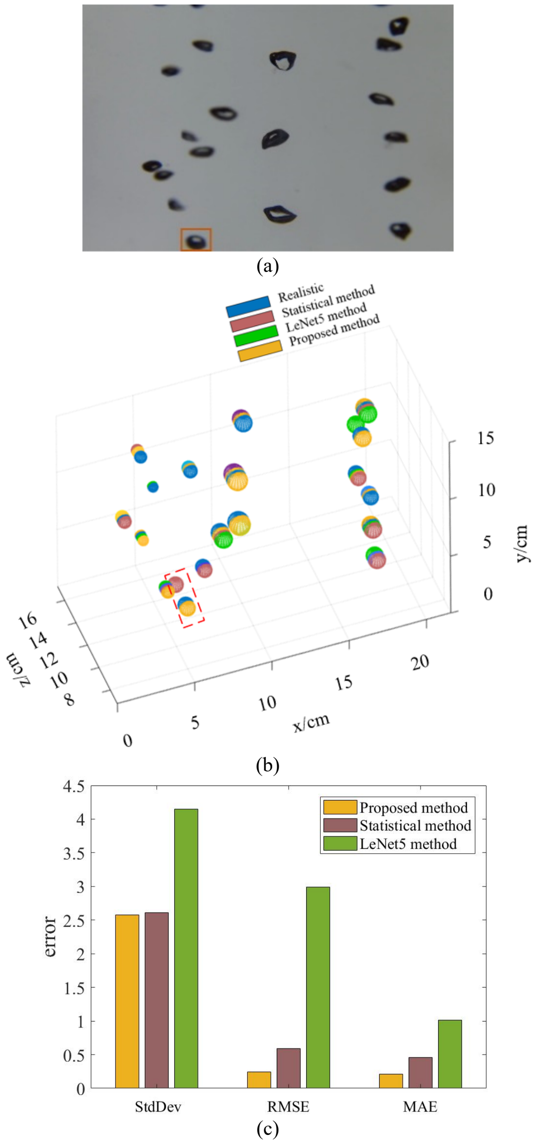

Fig. 17. 3D reconstruction result based on different method. (a) Refocused bubble image for reconstruction; (b) Contrast of 3D reconstructed bubble field; (c) Error of 3D reconstructed bubble field.

Fig. 18 is the reconstruction result comparison for the bubble without focused image in the refocused image sequence. Fig. 18(a) shows a bubble without focused image in the refocused image sequence. Fig. 18(b) compares the reconstructed results based on different methods and the ground truth position. Fig. 18(c) is the ground truth position of the bubble in three-dimensional space. Fig. 18(d) shows the reconstructed result based on statistical method, which is of obvious error. Since there is no focused image in this bubble image sequence, the statistical method could not acquire the accurate bubble depth by comparing the sharpness of bubble images. As can be seen in Fig. 18(e), there is no reconstructed result by LeNet5 method, because LeNet5 could not calculate the bubble depth by classification without bubble focused image. Fig. 18(f) is the reconstructed result based on DIF-LeNet with the refocused image of Fig. 18(a), which much closer to the ground truth position than the statistical method. The experiment results have strongly verified that the reconstruction performance of the proposed method is outstanding, especially for the bubble without focused image.

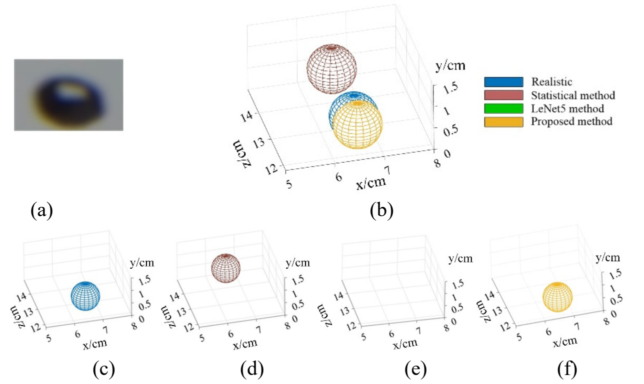

Fig. 18. Contrast of the reconstruction results for bubble without focused image based on different reconstruction methods. (a) Bubble image; (b) The reconstruction results of different reconstruction methods; (c) The real position of bubble; (d) Reconstruction result based on statistical method; (e) Reconstruction result based on LeNet5; (f) Reconstruction result based on DIF-LeNet.

The mean accuracy and the time consuming of 3D reconstruction for the bubbles in second-processed 3D reconstruction verification image set based on different reconstruction methods are contrasted in TABLE IV. As can be seen, the mean accuracy of the DIF-LeNet is higher than the statistic method and LeNet5 method. Because the DIF-LeNet can not only accurately obtain the depth information of ordinary bubbles, but also accurately obtain the depth of the edge bubble without focused bubble image.

TABLE IV
THE ACCURACY AND TIME CONSUMING OF 3D RECONSTRUCTION METHODS

Method	Time consuming of model training (s)	Time consuming of 3D reconstruction (s/bubble)	Mean accuracy of 3D reconstruction (%)
Statistical	/	11.6226	95.8
LeNet5	13430	8.4527	90.4
DIF-LeNet (proposed)	14709	1.8899	96.1

Although the training time of DIF-LeNet model is longer than LeNet5 model for the same training samples, the training time does not affect the reconstruction efficiency because of that the optimal parameters of the model are fixed once the training process is finished. In addition, the training time of the model is closely related to the number of training data, the structure of the model and the configuration of the computer. In the work of 3D reconstruction, the time consuming of test is a reasonable index to evaluate the reconstruction speed. As shown by the time consuming of 3D reconstruction in TABLE IV, the reconstruction speed of the DIF-LeNet model is faster than the other two methods. The reason for the reducing of reconstruction computation is that the DIF-LeNet model calculates the depth only based on one refocused light field image for each bubble and the other two methods have to compute and compare hundreds of images to search the bubble depth.

V. CONCLUSION

This paper provides a novel method for the reconstruction of 3D bubble field based on light field images. An improved deep learning network called DIF-LeNet is proposed for bubble depth prediction. The 3D reconstruction process of bubble field based on DIF-LeNet is detailed in this paper. To enhance the bubble characteristics, the DIF-LeNet fuses the bubble refocused image with its focal distance, which realizes the data fusion of different dimension. In order to compute the bubble depth, the last connection layer of DIF-LeNet is designed as regression network. Comparing with the reconstruction results based on statistical method and the LeNet5 method, both the accuracy and the speed of 3D reconstruction based on DIF-LeNet is improved. By the DIF-LeNet model, the position of any bubble in the refocused image could be predicted as long as the focal distance of the refocused image is known. However, it is necessary to pointed out that this method is not suitable for dense bubbles or overlapped bubbles. Additionally, the bubble shape is represented by a sphere of equal equivalent radius, which needs further improvements in the future work. The proposed DIF-LeNet provides a new idea for the data fusion method of different dimensions in the neural network, which has broad application prospects.

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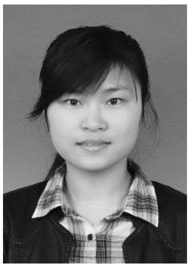
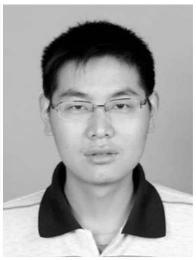


FIRST A. HONGYI WANG received the B.S. degree from the School of Electrical Engineering, Yanshan University, Qinhuangdao, China, in 2006, and the M.S. and the Ph. D degrees in School of Electrical Engineering and Automation from Tianjin University, Tianjin, china, both in 2011.

She is currently an associate professor in Tiangong University. Her main research interests include multi-phase flow measurement, intelligent detection technology, pattern recognition and fault diagnosis.

SECOND B. HONGYU WANG received the B.S. degree from Yanshan University, Hebei, China, in 2017. She is currently working toward the M.S. degree in the school of electrical engineering and automation, Tianjin Polytechnic University, TianJin, China.

She is a Research Member of Key Laboratory of Advanced Electrical Engineering and Energy Technology. Her research interests include pattern recognition, deep learning, and intelligent detection technology.

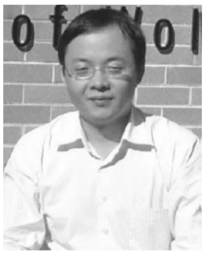


THIRD C. XINJUN ZHU received the B.S. degree from Linyi University, in 2008, the M.S. degree from the Shandong University of Technology, in 2011, and the Ph.D. degree from Tianjin University, in 2015. He joined Tiangong University, in 2015.

His research interests include fringe analysis, phase retrieval for optical 3-D measurement, and light scattering techniques for particles.

FOURTH D. LIMEI SONG received the B.S., M.S. and Ph.D. degrees from Tianjin University, Tianjin, China, in 1999, 2001, and 2004, respectively, all in measurement technology and instrumentation. She was an Associate Professor with the Southwest University of Science Technology from 2006 to 2008. She was a Visiting Scholar with the Hiroshima Institute of Technology, Japan, in 2006 and with the University of Wollongong, Wollongong, Australia from 2011 to 2012. In 2008, she joined as an Associate Professor with Tianjin Polytechnic University, where she is currently a Professor and the Vice Dean of the School of Electrical engineering and Automation.

Her research interests include machine vision, 3-D optical measurement, reverse engineering, and pattern recognition.



FIFTH E. QINGHUA GUO (S'07-M'08-SM'18) received the B.S. degree in electronic engineering and the M.S. degree in signal and information processing from Xidian University, Xi'an, China, in 2001 and 2004, respectively, and the Ph.D. degree in electronic engineering from the City University of Hong Kong, Hong Kong, in 2008. He is currently with the School of Electrical, Computer and Telecommunications Engineering, University of Wollongong, Wollongong, NSW, Australia.

His research interests include signal processing, 3-D reconstruction, and telecommunications. He was a recipient of the Australian Research Council's Discovery Early Career Researcher Award.

SIXTH F. FENG DONG (M'03-SM'06) received his B.S., M.S., and Ph.D. degrees in control science and engineering from Tianjin University, Tianjin, China, in 1988, 1996, and 2002, respectively. He is a professor with the School of Electrical and Information Engineering, Tianjin University; and the director of Tianjin Key Laboratory of Process Measurement and Control.

His research interests include process parameter detection and control system, industrial process tomography, multiphase flow measurement, and intelligent information processing.